Alexander Shashkov

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Education

University of California, Berkeley 2025-
PhD in Mathematics
Advisor: Sug Woo Shin

University of Cambridge 2024-2025
Master of Advanced Studies in Pure Mathematics, graduated with Distinction

Williams College 2019-2024
BA in Mathematics and Computer Science, GPA 3.94/4.00
Magna Cum Laude, Highest Honors in Mathematics

Publications

1. **Low lying zeros of Rankin-Selberg L-functions**. *Journal of Number Theory* (2026).
2. **Moments of one-level densities in families of cusp forms in the level aspect** (with Peter Cohen, Justine Dell, Oscar E. González, Simran Khunger, Chung-Hang Kwan, Steven J. Miller, Alicia Smith Reina, Carsten Sprunger, Nicholas Triantafillou, Nhi Truong, Roger Van Peski, Stephen Willis). *Algebra & Number Theory* (2026).
3. **Modular forms and an explicit Chebotarev variant of the Brun-Titchmarsh theorem** (with Daniel Hu & Hari Iyer). *Research in Number Theory* (2023).
4. **Adversarial agent-learning for cybersecurity: a comparison of algorithms** (with Erik Hemberg, Miguel Tulla, & Una-May O'Reilly). *The Knowledge Engineering Review* (2023).
5. **Limiting Spectral Distributions of Families of Block Matrix Ensembles** (with Teresa Dunn, Henry L. Fleischmann, Faye Jackson, Simran Khunger, Steven J. Miller, Luke Reifenberg, & Stephen Willis). *PUMP Journal of Undergraduate Research* (2022).
6. **Self-similar sets with arbitrary Hausdorff and box-counting dimension**. *The Pi Mu Epsilon Journal* (2021).
7. **Analyzing Student Reflection Sentiments and Problem-Solving Procedures in MOOCs** (with Robert Gold, Erik Hemberg, ByeongJo Kong, Ana Bell, & Una-May O'Reilly). *Proceedings of the Eighth ACM Conference on Learning @ Scale* (2021).

Research Experience

Williams College September 2022 – August 2023
Undergraduate thesis work in analytic number theory under Steven J. Miller.

University of Virginia June 2022 – July 2022
Research in analytic number theory under Ken Ono and Jesse Thorner.

Williams College June 2021 – August 2021
Research in analytic number theory and random matrix theory under Steven J. Miller.

MIT Computer Science and Artificial Intelligence Lab June 2020 – June 2021
Research in applied machine learning under Erik Hemberg and Una-May O'Reilly.

Awards

NSF GRFP Honorable Mention April 2025
Rosenburg Prize in Mathematics, Williams College Math Dept. June 2024
Phi Beta Kappa, Williams College June 2024
Sigma Xi, Williams College June 2024
Churchill Scholarship December 2023
Goldwater Scholarship April 2023

Talks

- *The Hodge-Tate decompositon via perfectoid spaces.* Berkeley p-adic Hodge theory learning seminar, October 2026.
- *Perfectoid spaces and the pro-étale site.* Berkeley p-adic Hodge theory learning seminar, September 2026.
- *The mod p local Langlands correspondence for* $\mathrm{GL}_2(\mathbb{Q}_p)$. Berkeley p-adic local Langlands learning seminar, September 2026.
- *Representations of reductive groups over local fields.* Berkeley student number theory seminar, October 2025.
- *Extending support for the centered moments of the low lying zeros of cuspidal newforms.* 34th Automorphic Forms Workshop, March 2022.
- *Extending support for the centered moments of the low lying zeros of cuspidal newforms* (with Simran Khunger). Maine-Quebec Number Theory Conference, October 2021.
- *Limiting Spectral Distributions of Families of Block Matrix Ensembles* (with Teresa Dunn, Henry Fleischmann, & Stephen Willis). Young Mathematicians Conference, August 2021.
- *Analyzing student reflection sentiments and problem-solving procedures in MOOCs.* Eighth ACM Conference on Learning @ Scale, June 2021.

Teaching & Outreach

Teaching Asst.	Math 10A Calculus	UC Berkeley	Spring 2026
Teaching Asst.	Math 54 Lin. Alg. & Diff. Eq.	UC Berkeley	Spring 2026
Teaching Asst.	Math 54 Lin. Alg. & Diff. Eq.	UC Berkeley	Fall 2025
Teaching Asst.	Math 383 Complex Analysis	Williams College	Fall 2023
Teaching Asst.	Math 409 Putnam Seminar	Williams College	Fall 2022
Teaching Asst.	Math 250 Linear Algebra	Williams College	Spring 2022
Referee	Journal of Number Theory		
Referee	PUMP Jour. of Undergrad Research		
Mentor	Prison Math Project		

Relevant Skills

Languages: English (native speaker), Russian (intermediate)
Programming: C++, Java, Python